UCLA Extension

405746: Large Language Models

COM SCI-X 450.46

Winter 2026 Section 1 4 Credits 01/05/2026 to 03/16/2026 Modified 12/29/2025

Contact Information

When needed, students are encouraged to contact the instructor through the Canvas Inbox (https://community.canvaslms.com/docs/DOC-10573-4212710324). Please allow up to **48 hours** for a response.

UCLA Extension Department Contact:

- Email: cs@uclaextension.edu

Instructor: Dr. Ben Winjum

Email: bwinjum@oarc.ucla.edu

Office Hours

Office hours via Zoom meeting are available **by appointment**. To schedule a meeting, please send a request via Canvas Inbox with several suggestions for days/times to meet.

Description

This course explores the theoretical concepts, design principles, and practical applications of large language models (LLMs). Students will gain a deep understanding of how LLMs process and generate human language, with a focus on both conceptual understanding and project-based application for real-world tasks. Topics covered include the transformer architecture, self-attention mechanisms, prompt engineering, model fine-tuning, and advanced applications like retrieval and agent systems. This course gives students practical experience implementing and customizing LLMs for applications such as text generation, classification, summarization, semantic search, and question answering, and they will work with both open-source and proprietary tools and frameworks, including LangChain, Hugging Face, and OpenAI. By the end of the course, students will be able to fine-tune pre-trained LLMs, evaluate their performance, and apply them effectively. Prior experience with Python is required; prior experience with deep learning will be beneficial.

Materials

Students must have a computer with internet access.

The following book is required:

- **Title:** Hands-On Large Language Models: Language Understanding and Generation
- **Authors:** Jay Alammar and Maarten Grootendorst
- **ISBN:** 1098150961

All other reading material will be provided online.

The course will also make extensive use of Python, language models, and online applications and technologies. Instructions regarding installation, access, and execution will be covered as needed.

Evaluation

Criteria

Graded Activities	Percentage	Notes
Quizzes	5%	Weekly short quizzes to enhance learning
Discussion Posts	10%	Weekly readings
Assignments	60%	Weekly coding assignments
Final Project	25%	Culminating project demonstrating concepts learned throughout course

Quizzes

Brief quizzes will be given each week to reinforce key course concepts and ensure consistent engagement. Each quiz will consist of 10-20 multiple-choice questions focused on recent material. These assessments are designed to encourage content review, gauge understanding, and provide timely feedback. Quizzes should be considered "open book" and are conducted without time limit.

Discussion Posts

Each week there will be an assigned reading(s), such as of an online tutorial/review or foundational article. I will post a prompt or question on Canvas, and you will be required to submit a short discussion post in response to it by Sunday. The posts should be thoughtful and substantive but relatively brief, just a few sentences up to two paragraphs max in length. These posts are intended to get you thinking about the material. You should also read and respond to the posts of your peers. You'll get one point for completing a post that is on-topic and one point for responding to at least one peer.

Assignments

Assignments will be the majority portion of your grade and consist of coding exercises, short answer questions that reflect on content, and preliminary work to get you moving smoothly towards your culminating final project.

These assignments will be distributed as Jupyter notebooks and will be graded according to the following criteria:

- Does it work? Unless you purposefully created code that produces errors to make a point, notebook assignments must run from top to bottom without any errors. Verify this by restarting the kernel and running all cells.
- Does it work correctly? It's possible to write code that runs ok but does not produce the intended results. The code should be bug-free.
- Cleanliness. Nobody wants to go through unreadable code! Make sure to document your work accordingly, providing markdown cells and comments throughout.
- Thinking out of the box. It is easy to copy an existing notebook and replace datasets and parameters to fulfill an assignment, but how well have you grasped the underlying concepts? This can be demonstrated by *applying* your skills in new ways rather than *copying* each step of a given assignment. For example, you may experiment with functions not demonstrated in class, or create your own workflow that borrows certain concepts learned in class to make them your own.

Final Project

Your final project will be the design and development of an LLM-based application. We will cover this more thoroughly during the 2nd and 3rd weeks when we have more exposure to LLM fundamentals. All necessary coding details will be covered over the course of our class.

Breakdown

Grade	**Range**
A+	100% - 97%
A	< 97% to 93%
A-	< 93% to 90%
B+	< 90% to 87%
B	< 87% to 83%
B-	< 83% to 80%

Grade	Range
C+	< 80% to 77%
C	< 77% to 73%
F	< 73%

Course Policies

Late Work Policy

- If you cannot submit something on time, please contact me before a due date and we can make alternative arrangements.
- If we do not make prior arrangements, I will give a one-day grace period on assignments, following which 5% will be deducted from your assignment grade for each subsequent day.
  - Ideally we should make alternative arrangements first before falling into this edge case.
- For final projects, the first prototype is due on Monday of Week 10 (3/9), feedback to peers is due that Friday (3/13) and the final version is due the following Monday (3/16).
  - These are not flexible deadlines unless you discuss with me.

Institutional Policies

Student Conduct

Students are subject to disciplinary action for several types of misconduct or attempted misconduct, including but not limited to academic dishonesty, such as cheating, multiple submission, plagiarism, unauthorized use of AI, or knowingly furnishing false information to the University; or behavioral misconduct, such as theft or misuse of the intellectual property of others, harassment, or disruption of the learning environment. Students are required to acknowledge the Extension Student Conduct Code and other applicable policies as a condition of enrollment. Please review the UCLA Extension Student Handbook (https://www.uclaextension.edu/sites/default/files/pdf/UNEX_Student_Handbook.pdf) and return your attestation form using the instructions on the final page. Report incidents of concern using the Incident Reporting Form (https://uclaextension.caseiq.app/portal).

Services for Students with Disabilities

In accordance with the Americans with Disabilities Act of 1990, UCLA Extension provides appropriate accommodations and support services to qualified applicants and students with disabilities. These include, but are not limited to, auxiliary aids/services such as sign language interpreters, assistive listening devices for hearing-impaired individuals, extended time for and proctoring of exams, and registration assistance. Accommodations and types of support services vary and are specifically designed to meet the disability-related needs of each student based on current, verifiable medical documentation. Arrangements for auxiliary aids/services are available only through UCLA Extension's

Service for Students with Disabilities Office at (310) 825-7851 or by email at ODS@uclaextension.edu. For complete information, please visit Accessibility and Disability Services (https://www.uclaextension.edu/enrollment-and-support/accessibility-and-disability-services).

Incompletes

The interim grade Incomplete (I) may be approved for a student who has completed the majority of the course requirements, with passing quality (grade C or higher), but is unable to complete a small portion of the coursework by the course end date for good cause. For courses in which an Incomplete may be allowed, approval by the instructor of record and the academic program director is required. The Incomplete grade is not an option for courses that do not bear credit, such as 700, 800, or 900-level courses.

- It is the student's responsibility to petition for an Incomplete by emailing the appropriate academic program department (https://www.uclaextension.edu/contact-ucla-extension) at least one week before the end of the course. The Program Department will initiate the petition process once the email is received.
- The student, the instructor, the CE/Program Director, and the program staff must complete the petition prior to the final course meeting or before the quarter end date (https://www.uclaextension.edu/calendar). This process can take up to one week to complete.
- The instructor will approve or deny the request. The instructor will provide details on what the student needs to accomplish in order to complete the course, as well as a due date for submitting completed work. The due date cannot exceed the end of the ensuing quarter (https://www.uclaextension.edu/calendar) when a final grade must be reported or the Incomplete lapses to the grade "F," "NP," or "U." Visit UCLA Extension Grading Scale (https://www.uclaextension.edu/transcripts-credits-grades/grading-scale) for more information.

An Incomplete allows the student to complete only work that is outstanding and does not allow prior completed work to be retaken or resubmitted.

All Grades are Final

No change of grade may be made by anyone other than the instructor, and then, only to correct clerical errors. No term grade except Incomplete may be revised by re-examination. The correction of a clerical error may be authorized only by the instructor of record communicating directly with personnel of Student and Alumni Services.

Sexual Harassment

The University of California is committed to creating and maintaining a community where all individuals who participate in University programs and activities can work and learn together in an atmosphere free of harassment, exploitation, or intimidation. Every member of the community should be aware that the University prohibits sexual harassment and sexual violence, and that such behavior violates both law and University policy. The University will respond promptly and effectively to reports of sexual harassment and sexual violence, and will take appropriate action to prevent, to correct, and when necessary, to discipline behavior that violates our policy.

All Extension students and instructors who believe they have been sexually harassed are encouraged to contact the Department of Student and Alumni Services for complaint resolution: UCLA Extension, 1145 Gayley Ave., Los Angeles, CA 90024; Voice/TTY: (310) 825-7031. For more information, please view the University's full Policy on Sexual Harassment and Sexual Violence (https://policy.ucop.edu/doc/4000385/SVSH).

Additional Items

Protecting Privacy and Data During Live Instruction

Live meeting sessions for this class, when applicable, are being conducted over Zoom. As the host, the instructor may be recording live sessions. Only the host has the ability to record meetings, no recording by other means is permitted. Recorded sessions will be posted in the Videos area of this class unless otherwise notified. Due to privacy, recordings are not available for download and are only accessible via Canvas for the duration of the class. If you have privacy concerns and do not wish to appear in the recording, do not turn on your video and/or audio. If you also prefer to use a pseudonym instead of your name, please let the instructor know what name you will be using so that the instructor knows who you are during the session. To rename yourself during a Zoom meeting, click on Participants, click on your name, click on More, click on Rename. If you would like to ask a question, you may do so privately through the Zoom chat by addressing your chat question to the instructor only (and not to ""everyone""). Additionally, chat may be used and moderated for live questions, and saving of chats is enabled. If you have questions or concerns about this, please contact the instructor via Canvas Inbox.

Pursuant to the terms of the agreement between Zoom and UCLA Extension, the data is used solely for this purpose and Zoom is prohibited from re-disclosing this information. UCLA Extension also does not use the data for any other purpose. Recordings will be deleted when no longer necessary. However, recordings may become part of an administrative disciplinary record if misconduct occurs during a video conference.

Course and Instructor Evaluation

UCLA Extension values your feedback on course and instructor evaluations. We ask all students to take a few minutes to complete an end-of-course evaluation survey. Updates to the course and instruction are influenced by your feedback. Understanding your student experience is essential to ensure continuing excellence in the online classroom and is appreciated by your instructor and the UCLA Extension academic leadership.

Your participation in a survey is voluntary, and your responses are confidential. After instructors submit grades, they will be given an evaluation report, but this report will not contain your name.

About Your Online Course Materials

Please note the following about online course components at UCLA Extension:

- Students must have basic computer skills, including the use of word processing software, email, and the ability to use internet browsers, such as Safari, Firefox, or Chrome.

- Students are responsible for meeting the technical requirements of Canvas and familiarizing themselves with the Canvas Learning Management System.
  - What are the browser and computer requirements for Canvas? (https://community.canvaslms.com/t5/Canvas-Basics-Guide/What-are-the-browser-and-computer-requirements-for-Instructure/ta-p/66)
- Students are responsible for keeping a copy of all assignments and work submitted, and to be aware of all assignments, due dates, and course guidelines.
- Students have access to courses via Canvas for an additional 30 calendar days after the course end date listed in the syllabus (the first 14 days are full access; the rest are read-only).
- Students are encouraged to download/print content throughout the duration of the course and before the additional 30-day access ends. No further access is possible after the course becomes unavailable.

  To download all your assignment submissions in Canvas, please refer to the online support guide. (https://community.canvaslms.com/docs/DOC-10606) for more information or contact Canvas Support via the help menu within Canvas.

UCLA Extension Canvas and Learning Support

For immediate 24/7 Canvas technical support, including holidays, click on **Help** (located on the menu to the left) where you can call or chat live with a Canvas Support representative.

UCLA Extension Academic Technology and Learning Innovation
The UCLA Extension Learning Support staff assists both students and instructors with Canvas-related technical support, as well as general and administrative questions.

Learning Support staff is available Monday through Friday, from 8 AM to 9 PM (Pacific Time), except holidays:

- Email: learningsupport@uclaextension.edu (mailto:learningsupport@uclaextension.edu)

Campus Safety Escorts

For students taking classes held on the UCLA campus and in and around Westwood Village, the UCLA Police Department provides a free walking escort service every day of the year from dusk until 1 a.m. Community Service Officers (CSOs) are available to walk students, faculty, staff members and visitors to and from anywhere on campus, in Westwood Village, and in the village apartments. CSOs are uniformed students who have received special training and are employed by the UCLA Police Department. To obtain an escort, please call (310) 794-9255 and allow 15 to 20 minutes for your escort to arrive. For complete information, see UCLA Evening Escorts (https://police.ucla.edu/cso/evening-escorts).

Schedule

When	Module Title	Notes

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Week 1	Foundations of Large Language Models (LLMs)	• History of language modeling, from statistical methods to transformers and LLMs • How LLMs work, common architectures, and comparing major model families • Running LLMs: Using Python with Hugging Face libraries and OpenAI API
Week 2	Transformer Architecture and Self-Attention	• Encoder-decoder structure, self-attention, positional encoding • Describing transformer architectures, pretraining strategies, and design motivations • Working with a basic transformer model using PyTorch
Week 3	Fine-Tuning and Adaptation	• Model specialization vs generalization • Pretraining on large corpora, transfer learning, zero-shot vs. fine-tuning • Fine-tuning an LLM for text classification
Week 4	Prompt Engineering	• The role of prompts in LLMs, designing effective prompts for different tasks • Zero-shot, few-shot, and chain-of-thought prompting • Experimenting with prompt engineering using OpenAI API or Hugging Face models
Week 5	Retrieval-Augmented Generation (RAG)	• RAG for knowledge retrieval • Embeddings, dense retrieval models, vector databases • Building a prototype RAG system for Q&A or summarization
Week 6	Agentic LLM Systems	• Linking applications and context with LLMs • ReAct, AutoGPT, and MCP • Building an AI agent using LangChain
Week 7	Evaluation, Risks, and Safety	• Evaluation frameworks: human evaluation, BLEU/ROUGE, GLUE, LLM-as-a-judge • Identifying and safeguarding against hallucinations, prompt injection, bias, misuse • Applying quantitative and qualitative evaluation methods
Week 8	Optimization and Approximation	• Model compression, distillation, quantization, pruning • Motivations behind approximations and trade-offs in size, speed, and accuracy • Compare trade-offs between raw and optimized models
Week 9	LLM Applications and Tasks	• Chatbots, semantic search, Q&A, summarization, code generation • Matching tasks to LLM design patterns and implementation strategies • Prototype real-world LLM use cases

When	Module Title	Notes
Week 10	Final Projects and Future Directions	• Building a practical LLM application • Peer review and feedback • Final reflections on emerging models, scaling, and next steps